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Section A: Concept Explanation

1. SRAM vs DRAM

SRAM uses flip flops to store data. It is faster, more expensive and is mainly used in cache memory.

DRAM uses capacitors to store data. It is slower, cheaper and is used in main memory

2. Locality of Reference

Locality of reference is the principal that programs access a small portion of memory repeatedly.

Its two types

Temporal locality - recently used data is likely to be used again.

Spatial locality - nearby memory locations are likely to accessed.

3. Cache Memory - Cache memory is used to reduce memory access time by storing frequently used data closer to the CPU. This improves overall processor performance

4. Cache indexing - The different cache indexing mechanism are
- Direct mapped
- Set associative
- Fully associative

5. Cache Block Size

A larger cache block size can improve performance by taking advantage of spatial locality but it can also increase miss penalty.

6. Cache Associativity

The no. of locations where a memory block can be placed in cache higher associativity reduces conflict misses.

7. Cache Size Tradeoff

A larger cache reduces miss rate, but it also increases cost and access latency.

8. <u>write policies</u>

<u>write-through</u> - updates both cache and main memory at every write

<u>write-back</u> - updates main memory only when the cache block is replaced.

9. Average Memory Access Time (AMAT)

AMAT is the average time required to access memory

AMAT = Hit Time + Miss Rate x Miss Penalty.

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10. Cache Replacement -

LRU stands for Least Recently Used. In this policy the block that has not been used for the longest time is replaced first.

11. Memory Interleaving

Memory interleaving is a technique in which multiple memory modules are accessed simultaneously to increase bandwidth.

12. Magnetic Disk

Tracks - are circular paths on the surface of the disk.

Sectors - are subdivisions of tracks that store data blocks.

13. Flash Memory -

Flash memory is a non-volatile memory used in SSDs and USB drives. It is faster than magnetic disks and has no moving parts.

14. Memory - Mapped I/O

In Memory mapped I/O devices have a seprate address space and are accessed using special instructions

15. I/O Mapped I/O - Devices have a seprate address space and are accessed using special instructions

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16. Programmed I/O - the CPU actively waits and controls I/O device operations.

17. Interrupt driven I/O - The device interrupts the CPU when it is ready, which improves CPU utilization.

18. DMA - DMA stands for Direct Memory Access. It allows devices to transfer data directly to memory without CPU intervention.

Section B:- Design Draw & Defend.

1. Memory Hierarchy Diagram

Registers
↓
Cache
↓
Main memory
↓
Secondary storage

Explanation - Registers are the fastest and smallest storage inside CPU. Cache is slower than registers but faster than main memory. Main Memory stores currently running programs and data. Secondary storage is the largest and

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2. Direct Mapped cache Organisation.

Main Memory Address

Tag	Index	offset

Cache :-

Index → selects cache line.
Tag → checks whether required block is present.
Offset → selects byte/word inside block.

Explanation- In direct mapping each main memory block maps to exactly one cache line The index selects the line the tag identifies the block and offset selects the required data within that block.

3. Write Through vs write back

Write - Through :-

CPU → Cache → Main Memory

↳ every write also goes to main memory.

Write - Back :-

CPU → cache
↳ main memory updated only when block is replaced.

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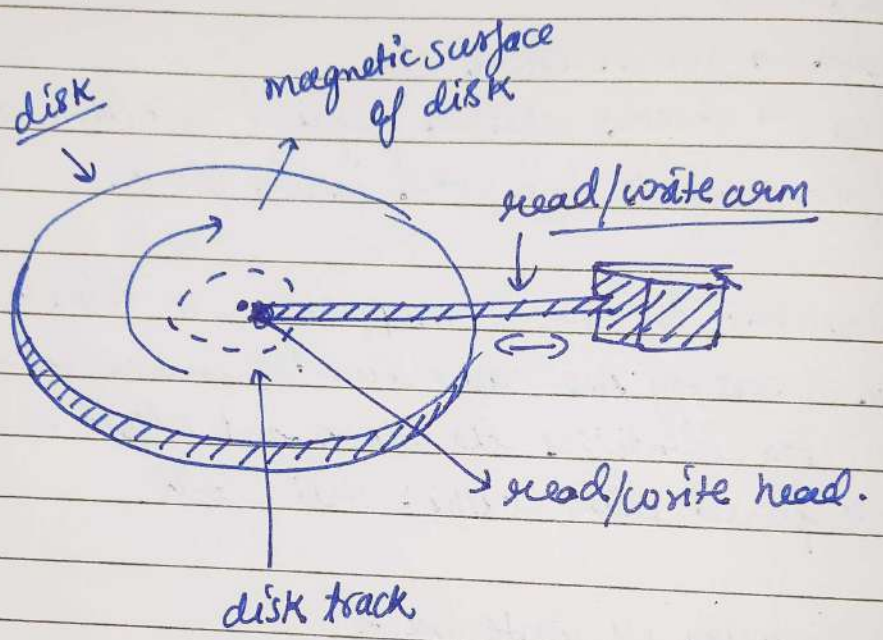
Explanation - Write through keeps memory always updated but increases memory traffic. Write-back reduces memory writes, but data in main memory may remain old until replacement occurs.

4. Magnetic Disk structure



a magnetic disk is made of circular tracks. Each track is divided into sectors, and each sector stores a block of data.

Section C :- Case-Based Analytical Questions.

1. A program repeatedly accesses the same memory location. Which locality principle applies and why?

This is temporal locality because the same memory location is being used again and again within a short time.

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2. Increasing cache block size reduces miss rate but sometimes slow performance why?

Because larger blocks increase miss penalty & may also cause pollution which can slow performance.

3. A system uses write-through cache. What happens during write operation?

During every write, the data is updated in both cache and main memory.

4. A high speed network card transfers large data blocks to memory without CPU involvement. Which technique is used?

The technique used is DMA (Direct Memory Access)

Section D:- One diagram + One Insight

CPU address

Tag | Index | offset

cache

Line 0	Tag	Data Block
Line 1	Tag	Data Block
Line 2	Tag	Data Block
Line 3	Tag	Data Block

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The CPU address is divided into tag index and offset

Index selects the cache line.

Tag is compared with stored tag

offset selects the required word/byte inside the block

2. AMAT Equation and Factors

AMAT = Hit Time + (Miss Rate × Miss Penalty)

Hit Time = Time to access cache when data is found.
Miss Rate = fraction of accesses that misses the cache.
Miss penalty = extra time needed to fetch data from lower memory

3. I/O data Transfer Techniques with one advantage

1. Programmed I/O

CPU ↔ I/O Device
CPU waits continously.

2. Interrupt-Driven I/O

CPU → Other work
I/O device → Interrupt → CPU

3. DMA

I/O Device ↔ Main memory
CPU only starts transfer.

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Section E:- Reflection, Assumption

1) write two assumptions you made while explaining cache operation

I assumed that the cache block size is fixed.
I assumed that the system uses a single-level cache.

2-) Mention one limitation of write through cache.

A limitation of write-through cache is increased memory traffic because every write goes to both cache and main memory.